**Note:** In homework assignments, when we are asked to "explain" or "justify", we need to provide a proof. In fact, when a math professor asks a question, they mean "provide the answer and prove that it is true." If they say "give a counterexample" they mean "give a counterexample and prove that it is a counterexample."

**Problem 1.** Let $A = \{\text{United States, Japan, China, India}\}$, $B = \{\text{England, Vietnam, India, Denmark}\}$ and $C = \{x \mid x \text{ is a country in Asia}\}$. List the elements of the following sets.

1. $A \cup B$
2. $(A \cap B) \backslash C$
3. $(B \cap C) \backslash A$

Are any of these disjoint from another (i.e., is their intersection empty)? Are any subsets of another?

**Solution.**

1. By definition,
$$\begin{aligned} A \cup B &= \{x \mid x \in A \text{ or } x \in B\} \\ &= \{\text{United States, Japan, China, India, England, Vietnam, Denmark}\}. \end{aligned}$$
2. Again, just by definition,
$$(A \cap B) \backslash C = \{x \mid x \in A \text{ and } x \in B \text{ and } x \notin C\} = \varnothing.$$
3. By definition,
$$(B \cap C) \backslash A = \{x \mid x \in B \text{ and } x \in C \text{ and } x \notin A\} = \{\text{Vietnam}\}.$$

Two sets $X$ and $Y$ are *disjoint* if $X \cap Y = \varnothing$. Notice that $\varnothing$ is both a subset of every set and also disjoint from every set. So yes, (2) is disjoint from both (1) and (3). Both (2) and (3) are subsets of (1) and (2) is also a subset of (3). □

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**Problem 2.** Each of the following sets is defined by the "roster method." Rewrite them in "set builder" notation.

1. $\{1, -2, 4, -8, 16, -32, \dots\}$
2. $\{\dots, -4, -1, 2, 5, 8, \dots\}$
3. $\{1, 3, 6, 10, 15, 21, 28, 36, \dots\}$
4. $\{-3, -2, -1, 0, 1, 2, 3, 4, 5\}$

**Solution.**

1. $\{1, -2, 4, -8, 16, -32, \dots\} = \{(-2)^n \mid n \in \mathbb{N} \cup \{0\}\}$
2. $\{\dots, -4, -1, 2, 5, 8, \dots\} = \{3n + 2 \mid n \in \mathbb{Z}\}$.
3. $\{1, 3, 6, 10, 15, 21, 28, 36, \dots\} = \left\{\frac{n(n+1)}{2} \mid n \in \mathbb{N}\right\}$.
4. $\{-3, -2, -1, 0, 1, 2, 3, 4, 5\} = \{x \in \mathbb{Z} \mid -3 \leqslant x \leqslant 5\}$.

□

**Problem 3.** 1. Express the empty set as the intersection of two non-empty sets, or explain why this is not possible.

2. Express the empty set as the difference of two non-empty sets, or explain why this is not possible.

3. Express the empty set as the union of two non-empty sets, or explain why this is not possible.

**Solution.**

1. One option is $\varnothing = \{1\} \cap \{2\}$.

2. One option is $\varnothing = \{1\} \setminus \{1\}$.

3. This is impossible. To justify this, we need to show that, for any sets $A$ and $B$, if $A \cup B = \varnothing$ then $A = \varnothing$ and $B = \varnothing$. *Proof.* Notice that if $x \in A$ or $x \in B$, then $x \in A \cup B$. Hence, $A \subseteq A \cup B$ and $B \subseteq A \cup B$. So if $\varnothing = A \cup B$, this says that $A \subseteq \varnothing$ and $B \subseteq \varnothing$. Therefore $A = B = \varnothing$. □

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**Problem 4.** We say that a subset $A$ of $\mathbb{R}$ is **closed under addition** if for all $x, y \in A$, it's always true that $x + y \in A$. For each of the following, either explain why the set is closed under addition or provide an example to show that it is not closed under addition.

1. The set of odd natural numbers.

2. The set of even natural numbers.

3. $\{\frac{1}{n^2} \mid n \in \mathbb{N}\}$.

**Solution.**

1. This set is not closed under addition. For example, 1 and 3 are in this set, but $1 + 3 = 4$ is not.

2. This set is closed under addition. Let $n, m$ be even numbers. By definition, $n = 2k$ and $m = 2\ell$ for some $k, \ell \in \mathbb{Z}$. Thus $n + m = 2k + 2\ell = 2(k + \ell)$. Since the integers are closed under addition, $k + \ell \in \mathbb{Z}$. Hence, by definition, $n + m$ is even.

3. This set is not closed under addition. For example, consider $\frac{1}{1} + \frac{1}{1} = \frac{2}{1}$. There is no way to write $\frac{2}{1} = \frac{1}{n^2}$ for $n \in \mathbb{N}$. Thus this set is not closed under addition. □

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**Problem 5.** For each of (a) through (d) below, do the following.

1. Write the statement as an English sentence that does not use the symbols $\forall$ and $\exists$.

2. Write the negation of the statement in symbolic form without using the symbol $\neg$.

3. Write the negation of the statement as an English sentence that does not use the symbols $\forall$ and $\exists$.

(Note: You do not need to ascertain the truth values for these statements.)

(a) $(\exists q \in \mathbb{Q})(q < 1/2)$.

**Solution.**

1. There exists a rational number $q$ such that $q < 1/2$.
2. $(\forall q \in \mathbb{Q})(q \geqslant 1/2)$.
3. For all rational numbers $q$, $q \geqslant 1/2$. □

(b) $(\forall n \in \mathbb{N})$(If $n = k^2$ for some $k \in \mathbb{N}$, then $2^n - 1$ is not prime.)

**Solution.**

1. For all natural numbers $n$, if $n = k^2$ for some $k \in \mathbb{N}$, then $2^n - 1$ is not prime.
2. $(\exists n \in \mathbb{N})(n = k^2$ for some $k \in \mathbb{N}$, but $2^n - 1$ is prime.)
3. There exists a natural number $n$ such that $n = k^2$ for some $k \in \mathbb{N}$, but $2^n - 1$ is prime. □

(c) $(\exists z \in \mathbb{Z})(\sqrt{2} < z < \sqrt{3})$.

**Solution.**

1. There exists an integer $z$ such that $\sqrt{2} < z < \sqrt{3}$.
2. $(\forall z \in \mathbb{Z})(z \leqslant \sqrt{2}$ or $z \geqslant \sqrt{3})$.
3. For all integers $z$, either $z \leqslant \sqrt{2}$ or $z \geqslant \sqrt{3}$. □

(d) $(\exists r \in \mathbb{R})(r^2 - r + 41 < 0)$.

**Solution.**

1. There exists a real number $r$ such that $r^2 - r + 41 < 0$.
2. $(\forall r \in \mathbb{R})(r^2 - r + 41 \geqslant 0)$.
3. For all real numbers $r$, $r^2 - r + 41 \geqslant 0$. □

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**Problem 6.** Write each of the following statements as an English sentence that does not use the symbols $\forall$ and $\exists$.

1. $(\exists m \in \mathbb{Z})(\forall n \in \mathbb{Z})(m > n)$.
2. $(\forall m \in \mathbb{Z})(\exists n \in \mathbb{Z})(m > n)$.
3. $(\forall m \in \mathbb{Z})(\forall n \in \mathbb{Z})(m > n)$.

**Solution.**

1. There exists an integer $m$ such that for all integers $n$, $m > n$. (I.e., there is some integer that is strictly larger than all integers.)
2. For all $m \in \mathbb{Z}$, there is some integer $n$ such that $m > n$. (I.e., given an integer $m$, there's some integer $n$ such that $m > n$.)
3. For all $m, n \in \mathbb{Z}$, $m > n$. □

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**Problem 7.** Write the negation of each statement from the previous problem in symbolic form without using the symbol $\neg$ *and* write each negation as an English sentence.

**Solution.**

1. Symbolic form: $(\forall m \in \mathbb{Z})(\exists n \in \mathbb{Z})(m \leqslant n)$
   Written English: For all $m \in \mathbb{Z}$, there exists some $n \in \mathbb{Z}$ such that $m \leqslant n$.
2. Symbolic form: $(\exists m \in \mathbb{Z})(\forall n \in \mathbb{Z})(m \leqslant n)$
   Written English: There exists an integer $m$ such that for all integers $n$, $m \leqslant n$.
3. Symbolic form: $(\exists m \in \mathbb{Z})(\exists n \in \mathbb{Z})(m \leqslant n)$
   Written English: There exist integers $m, n$ such that $m \leqslant n$. □

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**Problem 8.** Each of the following statements is false. Give a counterexample for each.

1. If $A, B$ are sets, then $(A \cup B)\backslash B = A$.
2. If $x \in \mathbb{Z}\backslash\{0\}$, then there exists a $y \in \mathbb{Z}$ such that $xy = 1$.
3. If $A, B, C$ are sets, then $(A \cup B)\backslash C = A \cup (B\backslash C)$.
4. If $n$ is a natural number, then there exist $a, b \in \mathbb{N}$ such that $a^2 + b^2 = n$.

**Solution.**

1. Let $A = B = \{1\}$. Then $(A \cup B)\backslash B = \varnothing \neq A$.
2. Let $x = 2$. We showed in class that 1 is not even, which means tat $2y = 1$ has no solution in $\mathbb{Z}$.
3. Let $A = C = \{1\}$ and $B = \{2\}$. Then $(A \cup B)\backslash C = \{2\}$ but $A \cup (B\backslash C) = \{1, 2\}$.
4. Recall that for us $0 \notin \mathbb{N}$ (the result is the same even if we consider 0 as a natural number, but now I can talk about a super simple example). Let $n = 1$. Then we know that $x^2 + y^2 = 1$ are all of the points on the unit circle. The only integer solutions are $(1, 0), (-1, 0), (0, 1), (0, -1)$. Since $0 \notin \mathbb{N}$, $n$ cannot be written as $n = x^2 + y^2$ where $x, y \in \mathbb{N}$. □

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**Problem 9.** For each of the following sets, write out (using logical symbols) what it means for an element $x$ to be an element of the set.

1. $(A\backslash B)\backslash C$
2. $A\backslash(B\backslash C)$
3. $(A\backslash B) \cup (A \cap C)$
4. $(A\backslash B) \cap (A\backslash C)$
5. $A\backslash(B \cup C)$

Then determine if any are equal by identifying equivalent expressions.

**Solution.**

1. Observe that $x \in (A\backslash B)\backslash C$ precisely when $(x \in A) \wedge (x \notin B) \wedge (x \notin C)$.
2. Observe that $x \in A\backslash(B\backslash C)$ precisely when

$$(x \in A) \wedge \neg[(x \in B) \wedge (x \notin C)] \equiv (x \in A) \wedge [(x \notin B) \vee (x \in C)].$$

3. Observe that $x \in (A \backslash B) \cup (A \cap C)$ precisely when

$$[(x \in A) \wedge (x \notin B)] \vee [(x \in A) \wedge (x \in C)] \equiv (x \in A) \wedge [(x \notin B) \vee (x \in C)].$$

4. Observe that $x \in (A \backslash B) \cap (A \backslash C)$ precisely when

$$[(x \in A) \wedge (x \notin B)] \wedge [(x \in A) \wedge (x \notin C)] \equiv (x \in A) \wedge (x \notin B) \wedge (x \notin C).$$

5. Observe that $x \in A \backslash (B \cup C)$ precisely when

$$[(x \in A) \wedge (x \notin B \cup C) \equiv (x \in A) \wedge (x \notin B) \wedge (x \notin C).$$

From our computations, it follows that $(A \backslash B) \backslash C = (A \backslash B) \cap (A \backslash C) = A \backslash (B \cup C)$, and also $A \backslash (B \backslash C) = (A \backslash B) \cup (A \cap C)$. □

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**Problem 10** (Practice.)**.** Do not turn these in, but I recommend completing the progress checks for sections 2.3 and 2.4 in your textbook, as well as the first bit of section 5.1